#This code will be used to demonstrate theorem 6.17. We will demonstrate that Q1=$\frac{x^2}{s^2}$ is decreasing along the γ1 curve; (***Part* 2 − *discarding potential critical points***)

#Define r1

$$
\begin{aligned}
r1 := \Big(& a^{32} b^{16} + b^{16} + \left(-4 b^{18} + 8 b^{16} - 4 b^{14}\right) a^{30} + \left(6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^{28} \\
&+ \left(72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}\right) a^{26} + \left(b^{32} + 8 b^{30} + 20 b^{28}\right. \\
&- 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\
&\left.- 157780 b^{12} - 32376 b^{10} + 1950 b^{8} + 1672 b^{6} + 20 b^{4} + 8 b^{2} + 1\right) a^{16} + \left(-1016 b^{26}\right. \\
&- 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\
&\left.+ 8 b^{8} + 8 b^{6}\right) a^{10} + \left(-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14}\right. \\
&\left.+ 144 b^{12} - 72 b^{10} - 9 b^{8}\right) a^{8} + \left(-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14}\right. \\
&\left.+ 72 b^{12}\right) a^{6} + \left(-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^{4} + \left(-4 b^{18} + 8 b^{16}\right. \\
&\left.- 4 b^{14}\right) a^{2} - 2560 \left(b^{2} - \frac{1}{5}\right) b^{7} \left(b^{2} - 2\right) \left(b^{2} + 1\right)^{4} a^{23} + 2 b^{4} \left(b^{24} + 36 b^{22} - 2636 b^{20}\right. \\
&- 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^{8} - 1372 b^{6} + 180 b^{4} \\
&\left.+ 36 b^{2} + 1\right) a^{12} - 4 b^{2} \left(b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16}\right. \\
&\left.+ 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^{8} + 378 b^{6} + 37 b^{4} + 12 b^{2} + 1\right) a^{14} - 4 b^{2} \left(b^{28}\right. \\
&+ 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\
&\left.+ 36360 b^{10} + 5211 b^{8} - 2118 b^{6} - 539 b^{4} + 76 b^{2} + 1\right) a^{18} + 2 b^{4} \left(b^{24} + 36 b^{22} + 180 b^{20}\right. \\
&- 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^{8} - 348 b^{6} + 3252 b^{4} \\
&\left.+ 420 b^{2} - 127\right) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^{8}\right. \\
&\left.+ 2369 b^{6} - 699 b^{4} + b^{2} + 1\right) b^{6} a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^{8} - 328 b^{6}\right. \\
&\left.- 528 b^{4} + \frac{4168}{9} b^{2} + 1\right) b^{8} a^{24} - 1024 b^{15} \left(b^{2} + 1\right)^{4} a^{7} - 2560 \left(b^{6} - \frac{8}{5} b^{4} + \frac{6}{5} b^{2}\right. \\
&\left.- 1\right) b^{11} \left(b^{2} + 1\right)^{4} a^{11} - 1024 \left(b^{2} - \frac{1}{2}\right) b^{13} \left(b^{2} + 1\right)^{5} a^{9} + 512 b^{7} \left(b^{10} - 19 b^{8} + 42 b^{6}\right. \\
&\left.- 9 b^{4} - 10 b^{2} + 1\right) \left(b^{2} + 1\right)^{4} a^{15} - 1536 \left(b^{8} - b^{6} + \frac{5}{3} b^{4} - \frac{16}{3} b^{2} + \frac{5}{3}\right) b^{9} \left(b^{2} + 1\right)^{4} a^{13}
\end{aligned}
$$

$$+ 512\, b^{7} \left(b^{10} - 22\, b^{8} + 81\, b^{6} - 78\, b^{4} + 11\, b^{2} + 1\right) \left(b^{2} + 1\right)^{4} a^{17} - 3584\, b^{7} \left(b^{8} - \frac{52}{7}\, b^{6} + \frac{95}{7}\, b^{4} - \frac{43}{7}\, b^{2} + \frac{3}{7}\right) \left(b^{2} + 1\right)^{4} a^{19} + 5632 \left(b^{6} - \frac{42}{11}\, b^{4} + \frac{38}{11}\, b^{2} - \frac{5}{11}\right) b^{7} \left(b^{2} + 1\right)^{4} a^{21}\Big) :$$

#The *candidate points*$\cdot$ (a, b) **for** *the intersection of r1* = 0 **and** $\tilde{\text{f}}$= 0 are of the form a \in A and b \in B where

$A := [3.192177604, 3.538902695, 3.555974116, 3.643966007, 3.884436541]$:
$B := [1.748533112, 3.464457702]$:

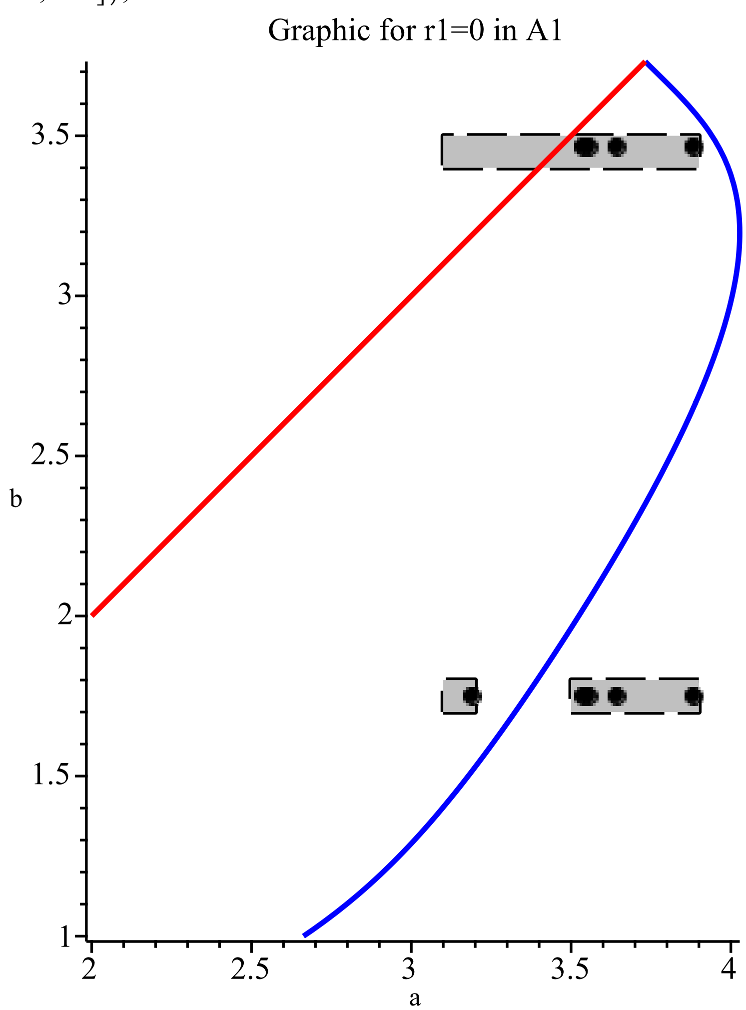
Points := [*seq*(*seq*([*a*, *b*], *b* **in** *B*), *a* **in** *A*)] :

Points_filtered := *select*(*p* → *p*[2] < *p*[1], *Points*);

Points_filtered := [[3.192177604, 1.748533112], [3.538902695, 1.748533112], [3.538902695, 3.464457702], [3.555974116, 1.748533112], [3.555974116, 3.464457702], [3.643966007, 1.748533112], [3.643966007, 3.464457702], [3.884436541, 1.748533112], [3.884436541, 3.464457702]] **(1)**

```
with(plots) :
with(plottools) :
r := a − b :


p1 := implicitplot(
   [r1 = 0, r = 0],
   a = 2 .. 5, b = 1 .. 2 + sqrt(3),
   scaling = constrained,
   gridrefine = 2,
   thickness = 2,
   color = [blue, red],
   labels = ["a", "b"],
    title = "Graphic for r1=0 in A1"
 ) :

# retangles
 R1 := rectangle([3.1, 1.7], [3.2, 1.8],
              color = silver, transparency = 0.5,
             linestyle = 3,   # pontilhado
             thickness = 2) :
  R2 := rectangle([3.5, 1.7], [3.9, 1.8],
              color = silver, transparency = 0.5,
             linestyle = 3,   # pontilhado
             thickness = 2) :
```

```
R3 := rectangle( [3.1, 3.4], [3.9, 3.5],
          color = silver, transparency = 0.5,
          linestyle = 3,   # pontilhado
          thickness = 2 ) :



# points
P := pointplot(
    Points_filtered ,
   symbol = solidcircle,
   symbolsize = 12,
    color = black
 ) :

# sobrepor tudo
display( [p1, P, R1, R2, R3 ]);
```



#By covering the previous points with rectangles, we will show that the curve r1=0 does not intersect them.

restart;

#Define r1

$$r1 := \Big(a^{32} b^{16} + b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} - 157780 b^{12} - 32376 b^{10} + 1950 b^{8} + 1672 b^{6} + 20 b^{4} + 8 b^{2} + 1) a^{16} + (-1016 b^{26} - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} + 8 b^{8} + 8 b^{6}) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} + 144 b^{12} - 72 b^{10} - 9 b^{8}) a^{8} + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} + 72 b^{12}) a^{6} + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{4} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{2} - 2560 \left(b^{2} - \frac{1}{5}\right) b^{7} (b^{2} - 2) (b^{2} + 1)^{4} a^{23} + 2 b^{4} (b^{24} + 36 b^{22} - 2636 b^{20} - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^{8} - 1372 b^{6} + 180 b^{4} + 36 b^{2} + 1) a^{12} - 4 b^{2} (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^{8} + 378 b^{6} + 37 b^{4} + 12 b^{2} + 1) a^{14} - 4 b^{2} (b^{28} + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} + 36360 b^{10} + 5211 b^{8} - 2118 b^{6} - 539 b^{4} + 76 b^{2} + 1) a^{18} + 2 b^{4} (b^{24} + 36 b^{22} + 180 b^{20} - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^{8} - 348 b^{6} + 3252 b^{4} + 420 b^{2} - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^{8} + 2369 b^{6} - 699 b^{4} + b^{2} + 1\right) b^{6} a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^{8} - 328 b^{6} - 528 b^{4} + \frac{4168}{9} b^{2} + 1\right) b^{8} a^{24} - 1024 b^{15} (b^{2} + 1)^{4} a^{7} - 2560 \left(b^{6} - \frac{8}{5} b^{4} + \frac{6}{5} b^{2} - 1\right) b^{11} (b^{2} + 1)^{4} a^{11} - 1024 \left(b^{2} - \frac{1}{2}\right) b^{13} (b^{2} + 1)^{5} a^{9} + 512 b^{7} (b^{10} - 19 b^{8} + 42 b^{6}$$

$$- 9\, b^{4} - 10\, b^{2} + 1 \big) \left(b^{2}+1\right)^{4} a^{15} - 1536 \left(b^{8} - b^{6} + \frac{5}{3}\, b^{4} - \frac{16}{3}\, b^{2} + \frac{5}{3}\right) b^{9} \left(b^{2}+1\right)^{4} a^{13}$$

$$+ 512\, b^{7} \left(b^{10} - 22\, b^{8} + 81\, b^{6} - 78\, b^{4} + 11\, b^{2} + 1\right) \left(b^{2}+1\right)^{4} a^{17} - 3584\, b^{7} \bigg(b^{8} - \frac{52}{7}\, b^{6}$$

$$+ \frac{95}{7}\, b^{4} - \frac{43}{7}\, b^{2} + \frac{3}{7}\bigg) \left(b^{2}+1\right)^{4} a^{19} + 5632 \left(b^{6} - \frac{42}{11}\, b^{4} + \frac{38}{11}\, b^{2} - \frac{5}{11}\right) b^{7} \big(b^{2}$$

$$+ 1\big)^{4} a^{21} \bigg) :$$

#Apply the Möbius transformation on the rectangle (3.1 , 3.2)× (1.7 , 1.8) that contains one point, constructing r̂

$$moebius := \left\{ a = \frac{\frac{31}{10} + \frac{32}{10} \cdot c}{1+c},\; b = \frac{\frac{17}{10} + \frac{18}{10} \cdot d}{1+d} \right\} :$$

r̂1 := collect(numer(expand(subs(moebius, r1))), c) :

#Compute the degree of r̂1 in c

degree(*r̂1*, *c*);

$$32 \tag{2}$$

#Define the coefficients ci = coeff (r̂1, c, i)

```
for i from 0 to 32 do
  c[i] := coeff(`r̂1`, c, i);
od:
```

#Compute the all degree of ci in d

seq(degree(c[i], d), i = 0 ..32);

32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32 **(3)**

#From here we will analyze the signal variation of all coefficients ci

$seq(seq(sign(coeff(c[j], d, i)), i = 32..0, -1), j = 32..0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, \tag{4}$$
$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$
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$-1, -1, -1, -1, -1$

restart

#Define r1

$$
\begin{aligned}
r1 := \Big(& a^{32} b^{16} + b^{16} + \left(-4 b^{18} + 8 b^{16} - 4 b^{14}\right) a^{30} + \left(6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^{28} \\
& + \left(72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}\right) a^{26} + \left(b^{32} + 8 b^{30} + 20 b^{28} \right.\\
& - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\
& \left. - 157780 b^{12} - 32376 b^{10} + 1950 b^{8} + 1672 b^{6} + 20 b^{4} + 8 b^{2} + 1\right) a^{16} + \left(-1016 b^{26} \right.\\
& - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\
& \left. + 8 b^{8} + 8 b^{6}\right) a^{10} + \left(-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \right.\\
& \left. + 144 b^{12} - 72 b^{10} - 9 b^{8}\right) a^{8} + \left(-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \right.\\
& \left. + 72 b^{12}\right) a^{6} + \left(-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^{4} + \left(-4 b^{18} + 8 b^{16} \right.\\
& \left. - 4 b^{14}\right) a^{2} - 2560 \left(b^{2} - \frac{1}{5}\right) b^{7} \left(b^{2} - 2\right) \left(b^{2} + 1\right)^{4} a^{23} + 2 b^{4} \left(b^{24} + 36 b^{22} - 2636 b^{20} \right.\\
& - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^{8} - 1372 b^{6} + 180 b^{4} \\
& \left. + 36 b^{2} + 1\right) a^{12} - 4 b^{2} \left(b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \right.\\
& \left. + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^{8} + 378 b^{6} + 37 b^{4} + 12 b^{2} + 1\right) a^{14} - 4 b^{2} \left(b^{28} \right.\\
& + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\
& \left. + 36360 b^{10} + 5211 b^{8} - 2118 b^{6} - 539 b^{4} + 76 b^{2} + 1\right) a^{18} + 2 b^{4} \left(b^{24} + 36 b^{22} + 180 b^{20} \right.\\
& - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^{8} - 348 b^{6} + 3252 b^{4} \\
& \left. + 420 b^{2} - 127\right) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^{8} \right.\\
& \left. + 2369 b^{6} - 699 b^{4} + b^{2} + 1\right) b^{6} a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^{8} - 328 b^{6} \right.\\
& \left. - 528 b^{4} + \frac{4168}{9} b^{2} + 1\right) b^{8} a^{24} - 1024 b^{15} \left(b^{2} + 1\right)^{4} a^{7} - 2560 \left(b^{6} - \frac{8}{5} b^{4} + \frac{6}{5} b^{2} \right.\\
& \left. - 1\right) b^{11} \left(b^{2} + 1\right)^{4} a^{11} - 1024 \left(b^{2} - \frac{1}{2}\right) b^{13} \left(b^{2} + 1\right)^{5} a^{9} + 512 b^{7} \left(b^{10} - 19 b^{8} + 42 b^{6} \right.\\
& \left. - 9 b^{4} - 10 b^{2} + 1\right) \left(b^{2} + 1\right)^{4} a^{15} - 1536 \left(b^{8} - b^{6} + \frac{5}{3} b^{4} - \frac{16}{3} b^{2} + \frac{5}{3}\right) b^{9} \left(b^{2} + 1\right)^{4} a^{13}
\end{aligned}
$$

$$+ 512\, b^7 \left(b^{10} - 22\, b^8 + 81\, b^6 - 78\, b^4 + 11\, b^2 + 1\right) \left(b^2 + 1\right)^4 a^{17} - 3584\, b^7 \left(b^8 - \frac{52}{7}\, b^6 + \frac{95}{7}\, b^4 - \frac{43}{7}\, b^2 + \frac{3}{7}\right) \left(b^2 + 1\right)^4 a^{19} + 5632 \left(b^6 - \frac{42}{11}\, b^4 + \frac{38}{11}\, b^2 - \frac{5}{11}\right) b^7 \left(b^2 + 1\right)^4 a^{21} \Big) :$$

#Apply the Möbius transformation on the rectangle $(3.5\,,\, 3.9) \times (1.7\,,\, 1.8)$ *that contains four points , constructing* $\hat{r1}$

$$moebius := \left\{ a = \frac{\frac{35}{10} + \frac{39}{10} \cdot c}{1 + c},\; b = \frac{\frac{17}{10} + \frac{18}{10} \cdot d}{1 + d} \right\} :$$

$\hat{r1} := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(\mathrm{moebius}, \mathrm{r1}))), c) :$

#Compute the degree of $\hat{r1}$ *in c*

$degree(\hat{r1}, c);$

$$32 \tag{5}$$

#Define the coefficients ci = coeff $(\hat{r1}, c, i)$

for *i* **from** 0 **to** 32 **do**
  $c[i] := coeff(\,`\hat{r1}`, c, i);$
od:

#Compute the all degree of ci in d
seq(degree(c[i], d), i = 0 ..32);

32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32 **(6)**

#From here we will analyze the signal variation of all coefficients ci

$seq(seq(sign(coeff(c[j], d, i)), i = 32 ..0, -1), j = 32 ..0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, **(7)**
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
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1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

restart

#Define r1

$$
\begin{aligned}
\mathrm{r1} := \Big(& a^{32} b^{16} + b^{16} + \left(-4 b^{18} + 8 b^{16} - 4 b^{14}\right) a^{30} + \left(6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^{28} \\
&+ \left(72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}\right) a^{26} + \left(b^{32} + 8 b^{30} + 20 b^{28} \right. \\
&- 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\
&\left. - 157780 b^{12} - 32376 b^{10} + 1950 b^{8} + 1672 b^{6} + 20 b^{4} + 8 b^{2} + 1\right) a^{16} + \left(-1016 b^{26} \right. \\
&- 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\
&\left. + 8 b^{8} + 8 b^{6}\right) a^{10} + \left(-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \right. \\
&\left. + 144 b^{12} - 72 b^{10} - 9 b^{8}\right) a^{8} + \left(-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \right. \\
&\left. + 72 b^{12}\right) a^{6} + \left(-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}\right) a^{4} + \left(-4 b^{18} + 8 b^{16} \right. \\
&\left. - 4 b^{14}\right) a^{2} - 2560 \left(b^{2} - \frac{1}{5}\right) b^{7} \left(b^{2} - 2\right) \left(b^{2} + 1\right)^{4} a^{23} + 2 b^{4} \left(b^{24} + 36 b^{22} - 2636 b^{20} \right. \\
&- 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^{8} - 1372 b^{6} + 180 b^{4} \\
&\left. + 36 b^{2} + 1\right) a^{12} - 4 b^{2} \left(b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \right. \\
&\left. + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^{8} + 378 b^{6} + 37 b^{4} + 12 b^{2} + 1\right) a^{14} - 4 b^{2} \left(b^{28} \right. \\
&+ 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\
&\left. + 36360 b^{10} + 5211 b^{8} - 2118 b^{6} - 539 b^{4} + 76 b^{2} + 1\right) a^{18} + 2 b^{4} \left(b^{24} + 36 b^{22} + 180 b^{20} \right. \\
&- 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^{8} - 348 b^{6} + 3252 b^{4} \\
&\left. + 420 b^{2} - 127\right) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^{8} \right. \\
&\left. + 2369 b^{6} - 699 b^{4} + b^{2} + 1\right) b^{6} a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^{8} - 328 b^{6} \right. \\
&\left. - 528 b^{4} + \frac{4168}{9} b^{2} + 1\right) b^{8} a^{24} - 1024 b^{15} \left(b^{2} + 1\right)^{4} a^{7} - 2560 \left(b^{6} - \frac{8}{5} b^{4} + \frac{6}{5} b^{2} \right. \\
&\left. - 1\right) b^{11} \left(b^{2} + 1\right)^{4} a^{11} - 1024 \left(b^{2} - \frac{1}{2}\right) b^{13} \left(b^{2} + 1\right)^{5} a^{9} + 512 b^{7} \left(b^{10} - 19 b^{8} + 42 b^{6} \right. \\
&\left. - 9 b^{4} - 10 b^{2} + 1\right) \left(b^{2} + 1\right)^{4} a^{15} - 1536 \left(b^{8} - b^{6} + \frac{5}{3} b^{4} - \frac{16}{3} b^{2} + \frac{5}{3}\right) b^{9} \left(b^{2} + 1\right)^{4} a^{13}
\end{aligned}
$$

$$+ 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} - 3584 b^7 \left(b^8 - \frac{52}{7} b^6 + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7}\right) (b^2 + 1)^4 a^{19} + 5632 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11}\right) b^7 (b^2 + 1)^4 a^{21}\Big) :$$

#Apply the Möbius transformation on the rectangle $(3.1\,,\,3.9) \times (3.4\,,\,3.5)$ *that contains four points , constructing* $\hat{r1}$

$$moebius := \left\{ a = \frac{\frac{31}{10} + \frac{39}{10} \cdot c}{1 + c}, b = \frac{\frac{34}{10} + \frac{35}{10} \cdot d}{1 + d} \right\} :$$

$\hat{r1}$:= collect(numer(expand(subs(moebius, r1))), c) :

#Compute the degree of $\hat{r1}$ *in c*

degree($\hat{r1}$, c);

32 **(8)**

#Define the coefficients ci = coeff($\hat{r1}$, c, i)

```
for i from 0 to 32 do
  c[i] := coeff(`r̂1`, c, i);
od:
```

#Compute the all degree of ci in d

seq(degree(c[i], d), i = 0 ..32);

32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32 **(9)**

#From here we will analyze the signal variation of all coefficients ci

seq(*seq*(*sign*(*coeff*(*c*[*j*], *d*, *i*)), *i* = 32 ..0, −1), *j* = 32 ..0, −1);

−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, **(10)**
−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1,
−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1,
−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1,
−1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1, −1,

$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$
$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$
$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$
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$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,$
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